

HULIN, China

WMO: 509830

Lat: 45.7606N

Lon: 132.8831E

Elev: 100

StdP: 100.13

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.8	-24.9	-30.5	0.2	-25.9	-28.7	0.3	-24.1	9.5	-13.2	8.6	-13.9	1.2	290	0.497

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	29.4	22.5	28.0	21.9	26.7	21.1	24.2	27.5	23.2	26.4	22.3	25.3	2.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	18.1	26.2	22.2	17.1	25.3	21.3	16.2	24.4	73.7	27.6	69.9	26.6	66.5	25.4	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.4	7.2	6.1	DB	-29.1	32.0	2.6	1.9	-31.0	33.4	-32.6	34.5	-34.1	35.6	-36.0	37.0	
			WB	-29.3	25.4	2.6	1.1	-31.2	26.2	-32.7	26.8	-34.2	27.4	-36.0	28.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-17.3	-12.8	-3.8	6.0	13.5	19.0	22.3	21.2	15.3	6.6	-5.2	-15.1
	DBStd	14.58	4.33	4.76	5.34	4.13	3.65	3.23	2.38	2.64	3.45	4.53	6.25	4.68
	HDD10.0	3419	848	639	428	132	11	0	0	0	3	124	455	779
	HDD18.3	5436	1106	872	687	369	154	30	2	7	101	364	705	1038
	CDD10.0	1310	0	0	0	13	118	271	381	347	162	18	0	0
	CDD18.3	285	0	0	0	0	4	51	124	96	11	0	0	0
	CDH23.3	1568	0	0	0	1	47	303	693	474	49	0	0	0
	CDH26.7	298	0	0	0	0	6	66	149	73	4	0	0	0
Wind	WSAvg	2.5	2.1	2.5	3.0	3.1	2.9	2.5	2.2	2.1	2.2	2.6	2.6	2.2
Precipitation	PrecAvg	556	5	7	12	34	53	81	104	116	77	43	18	8
	PrecMax	849	13	22	44	88	115	192	196	214	231	112	46	22
	PrecMin	358	0	0	1	6	19	20	28	16	11	1	3	0
	PrecStd	118	3	6	9	17	23	44	46	53	46	30	11	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.7	3.1	11.2	22.0	27.2	30.8	31.6	30.0	26.9	20.7	11.6	1.7
		MCWB	-3.8	0.1	5.5	11.5	16.7	21.3	23.2	24.0	20.4	13.4	8.0	0.3
	2%	DB	-6.8	0.4	8.1	18.7	24.5	28.5	29.7	28.6	24.6	18.1	8.3	-2.6
		MCWB	-8.3	-1.8	3.3	9.9	14.9	20.1	23.3	23.1	18.8	12.6	5.4	-4.0
	5%	DB	-8.9	-2.3	6.0	15.7	22.3	26.6	28.3	27.3	22.9	15.7	5.9	-5.5
		MCWB	-10.0	-4.2	2.0	8.4	14.0	19.3	22.5	22.1	17.5	10.9	3.3	-6.7
	10%	DB	-10.9	-5.1	3.6	13.2	20.4	24.8	26.9	26.0	21.4	13.7	3.4	-8.1
		MCWB	-11.9	-6.8	0.4	7.1	13.2	18.8	21.9	21.3	16.8	9.8	1.2	-9.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.7	1.0	6.4	12.3	18.0	23.2	25.9	25.1	21.9	15.3	8.7	0.5
		MCDB	-2.8	2.8	10.6	19.8	25.2	27.9	29.2	28.3	24.8	18.9	11.4	1.5
	2%	WB	-8.2	-1.8	3.9	10.7	16.2	21.7	24.4	24.1	20.3	13.6	5.8	-3.8
		MCDB	-6.9	0.4	7.4	17.6	22.4	26.4	27.9	27.2	23.1	16.4	7.7	-2.7
	5%	WB	-10.0	-4.2	2.1	9.1	15.1	20.6	23.5	23.2	18.9	12.0	3.5	-6.7
		MCDB	-8.9	-2.4	5.1	14.5	20.7	24.9	26.8	26.1	22.1	14.8	5.7	-5.6
	10%	WB	-11.9	-6.8	0.7	7.6	14.0	19.5	22.6	22.3	17.5	10.3	1.3	-9.3
		MCDB	-10.9	-5.2	3.4	12.4	18.6	23.6	25.7	24.9	20.5	13.3	3.2	-8.1
Mean Daily Temperature Range	5% DB	MDBR	8.6	9.6	8.8	9.6	9.3	8.1	7.2	7.7	10.0	9.7	8.0	7.8
		MCDBR	9.8	10.6	10.6	13.6	12.9	10.7	9.4	9.2	11.4	12.4	10.0	9.6
		MCWBR	8.7	8.8	6.8	7.2	5.7	4.5	4.3	4.3	5.8	7.3	7.4	8.7
	5% WB	MCDBR	9.9	10.3	9.9	12.4	11.1	9.4	8.0	7.7	9.3	10.8	9.5	9.6
		MCWBR	8.8	8.8	6.7	7.1	5.5	4.5	4.2	4.1	5.2	7.3	7.6	8.9
Clear-Sky Solar Irradiance	taub	0.310	0.354	0.433	0.547	0.573	0.552	0.512	0.474	0.429	0.447	0.380	0.325	
	taud	2.079	1.945	1.842	1.730	1.748	1.888	2.036	2.130	2.160	1.962	2.034	2.047	
	Ebn at Noon	756	786	771	718	718	737	763	778	777	673	653	680	
	Edh at Noon	97	137	177	217	220	192	164	144	128	135	103	91	
All-Sky Solar Radiation	RadAvg	2.11	3.25	4.40	4.72	5.26	5.61	5.10	4.70	4.08	2.85	1.96	1.65	
	RadStd	0.10	0.14	0.28	0.48	0.41	0.66	0.50	0.34	0.35	0.20	0.14	0.09	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page